



Introduction to HTML5:

Part 2

Outline

- Build a form using the modern HTML5 input types.
 - Use self-validating input elements.
 - Specify temporary placeholder text.
- Use autocomplete input elements that help users re-enter text that they've previously entered in a form.
- Use a datalist to specify a list of values.
- Use HTML5's new page-structure elements to delineate parts of a page.

3.1 New HTML5 Form input Types

- ▶ Figure 3.1 demonstrates HTML5's new form input types.
- ▶ These are not yet universally supported by all browsers.



```
1  <!DOCTYPE html>
2
3  <!-- Fig. 3.1: newforminputtypes.html -->
4  <!-- New HTML5 form input types and attributes. -->
5  <html>
6      <head>
7          <meta charset="utf-8">
8          <title>New HTML5 Input Types</title>
9      </head>
10
11     <body>
12         <h1>New HTML5 Input Types Demo</h1>
13         <p>This form demonstrates the new HTML5 input types
14             and the placeholder, required and autofocus attributes.
15         </p>
16
17         <form method = "post" action = "http://www.deitel.com">
18             <p>
19                 <label>Color:
20                     <input type = "color" autofocus />
21                     (Hexadecimal code such as #ADD8E6)
22                 </label>
23             </p>
```

Fig. 3.1 | New HTML5 form input types and attributes. (Part I of 5.)

```
24 <p>
25   <label>Date:
26     <input type = "date" />
27       (yyyy-mm-dd)
28   </label>
29 </p>
30 <p>
31   <label>Datetime:
32     <input type = "datetime" />
33       (yyyy-mm-ddThh:mm+ff:gg, such as 2012-01-27T03:15)
34   </label>
35 </p>
36 <p>
37   <label>Datetime-local:
38     <input type = "datetime-local" />
39       (yyyy-mm-ddThh:mm, such as 2012-01-27T03:15)
40   </label>
41 </p>
42 <p>
43   <label>Email:
44     <input type = "email" placeholder = "name@domain.com"
45       required /> (name@domain.com)
46   </label>
47 </p>
```

Fig. 3.1 | New HTML5 form input types and attributes. (Part 2 of 5.)

```
48 <p>
49   <label>Month:
50     <input type = "month" /> (yyyy-mm)
51   </label>
52 </p>
53 <p>
54   <label>Number:
55     <input type = "number"
56       min = "0"
57       max = "7"
58       step = "1"
59       value = "4" />
60   </label> (Enter a number between 0 and 7)
61 </p>
62 <p>
63   <label>Range:
64     0 <input type = "range"
65       min = "0"
66       max = "20"
67       value = "10" /> 20
68   </label>
69 </p>
```

Fig. 3.1 | New HTML5 form input types and attributes. (Part 3 of 5.)

```
70 <p>
71   <label>Search:
72     <input type = "search" placeholder = "search query" />
73   </label> (Enter your search query here.)
74 </p>
75 <p>
76   <label>Tel:
77     <input type = "tel" placeholder = "(###) ###-####"
78       pattern = "\(\d{3}\) +\d{3}-\d{4}" required />
79     (###) ###-####
80   </label>
81 </p>
82 <p>
83   <label>Time:
84     <input type = "time" /> (hh:mm:ss.ff)
85   </label>
86 </p>
87 <p>
88   <label>URL:
89     <input type = "url"
90       placeholder = "http://www.domainname.com" />
91     (http://www.domainname.com)
92   </label>
93 </p>
```

regular expression

Fig. 3.1 | New HTML5 form input types and attributes. (Part 4 of 5.)

https://www.w3schools.com/tags/att_input_pattern.asp

```
94      <p>
95          <label>Week:
96              <input type = "week" />
97                  (yyyy-Wnn, such as 2012-W01)
98          </label>
99      </p>
100     <p>
101         <input type = "submit" value = "Submit" />
102         <input type = "reset" value = "Clear" />
103     </p>
104 </form>
105 </body>
106 </html>
```

Fig. 3.1 | New HTML5 form input types and attributes. (Part 5 of 5.)

Regular Expression:

- https://www.w3schools.com/js/js_regexp.asp
- https://developer.mozilla.org/en-US/docs/Web/JavaScript/Guide/Regular_Expressions
- <https://regex101.com/>



New HTML5 Input Types Demo

This form demonstrates the new HTML5 input types and the placeholder, required and autofocus attributes.

Color: (Hexadecimal code such as #ADD8E6)

Date: (yyyy-mm-dd)

Datetime: (yyyy-mm-ddThh:mm+ff:gg, such as 2012-01-27T03:15)

Datetime-local: (yyyy-mm-ddThh:mm, such as 2012-01-27T03:15)

Email: (name@domain.com)

Month: (yyyy-mm)

Number: (Enter a number between 0 and 7)

Range: 0 20

Search: (Enter your search query here.)

Tel: (###) ###-####

Time: (hh:mm)

URL: (http://www.domainname.com)

Week: (yyyy-Wnn, such as 2012-W01)

3.1.1 `input` Type `color`

- ▶ The `color input type` enables the user to enter a color.
- ▶ At the time of this writing, most browsers render the `color input type` as a text field in which the user can enter a hexadecimal code or a color name.
- ▶ In the future, when you click a color input, *browsers will likely display a `color picker`* similar to the Microsoft Windows color dialog shown in Fig. 3.2.

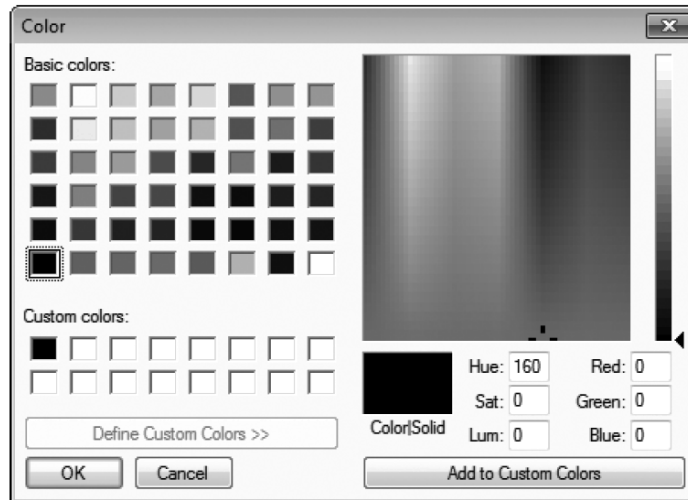


Fig. 3.2 | A dialog for choosing colors.

3.1.1 `input` Type `color`

autofocus Attribute

- ▶ The **`autofocus` attribute**—an optional attribute that can be used in only one `input` element on a form—automatically gives the focus to the `input` element, allowing the user to begin typing in that element immediately.
- ▶ Figure 3.3 shows autofocus on the `color` element—the first `input` element in our form—as rendered in Chrome. You do not need to include `autofocus` in your forms.

3.1.1 `input` Type `color` (cont.)

Validation

- ▶ The new HTML5 `input` types are *self validating* on the client side, eliminating the need to add complicated JavaScript code to your web pages to validate user input, reducing the amount of invalid data submitted and consequently reducing Internet traffic between the server and the client to correct invalid input.
- ▶ *The server should still validate all user input.*
- ▶ When a user enters data into a form then submits the form the browser immediately checks the self-validating elements to ensure that the data is correct (Fig. 3.4).

3.1.1 `input` Type `color` (cont.)

- ▶ Figure 3.5 lists each of the new HTML5 `input` types and provides examples of the proper formats required for each type of data to be valid.

3.1.1 input Type color (cont.)

- ▶ If you want to bypass validation, you can add the **formnovalidate** attribute to input type submit in line 101:

```
<input type = "submit" value = "Submit"  
formnovalidate />
```

3.1.2 `input` Type `date`

- ▶ The `date input type` enables the user to enter a date in the form `yyyy-mm-dd`.
- ▶ Firefox and Internet Explorer display a text field in which a user can enter a date such as `2012-01-27`.
- ▶ Chrome and Safari display a **spinner control** —a text field with an up-down arrow (▾) on the right side—allowing the user to select a date by clicking the up or down arrow.
- ▶ The start date is the *current date*.
- ▶ Opera displays a calendar from which you can choose a date.
- ▶ In the future, when the user clicks a date input, browsers are likely to display a date control similar to the Microsoft Windows one shown in Fig. 3.6.
- ▶ (Chrome已實作此功能)



年 / 月 / 日  (yyyy-mm-dd)

2025年09月 ▼

↑ ↓

日	一	二	三	四	五	六
31	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	1	2	3	4
5	6	7	8	9	10	11

清除 今天



3.1.4 `input Type datetime-local`

- ▶ The `datetime-local input type` enables the user to enter the date and time in a single control.
- ▶ The data is entered as year, month, day, hour, minute, second and fraction of a second.



年 / 月 / 日 -- -- : --  (yyyy-mm-ddThh:mm, such

2025年09月 ▾

↑ ↓

日	一	二	三	四	五	六
31	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	1	2	3	4
5	6	7	8	9	10	11

清除 今天

10 59 上午

11 00 下午

12 01

01 02

02 03

03 04

04 05

3.1.5 `input` Type `email`

- ▶ The `email` input type enables the user to enter an e-mail address or a list of e-mail addresses separated by commas (if the `multiple` attribute is specified).
- ▶ Currently, all of the browsers display a text field.
- ▶ If the user enters an *invalid* e-mail address (i.e., the text entered is *not* in the proper format) and clicks the Submit button, a callout asking the user to enter an e-mail address is rendered pointing to the input element (Fig. 3.7).
- ▶ HTML5 does not check whether an e-mail address entered by the user actually exists—rather **it just validates that the e-mail address is in the *proper format*.**

3.1.5 `input` Type `email` (cont.)

placeholder Attribute

- ▶ The `placeholder` attribute allows you to place temporary text in a text field.
- ▶ Generally, `placeholder` text is *light gray* and provides an example of the text and/or text format the user should enter (Fig. 3.8).
- ▶ When the *focus* is placed in the text field (i.e., the cursor is in the text field), the `placeholder` text disappears—it's not “submitted” when the user clicks the Submit button (unless the user types the same text).

3.1.5 `input` Type `email` (cont.)

- ▶ HTML5 supports placeholder text for only six input types—text, search, url, tel, email and password.

required Attribute

- ▶ The **required** attribute forces the user to enter a **value** before submitting the form.
- ▶ You can add `required` to any of the input types.
- ▶ In this example, the user *must* enter an e-mail address and a telephone number to submit the form (Fig. 3.9).

3.1.6 `input` Type `month`

- ▶ The `month input type` enables the user to enter a year and month in the format `yyyy-mm`, such as `2012-01`.
- ▶ If the user enters the data in an improper format (e.g., `January 2012`) and submits the form, a callout stating that an invalid value was entered appears.

3.1.7 `input` Type `number`

- ▶ The `number input type` enables the user to enter a numerical value—mobile browsers typically display a numeric keypad for this `input` type.
 - The `min` attribute sets the minimum valid number.
 - The `max` attribute sets the maximum valid number.
 - The `step` attribute determines the **increment** in which the numbers increase.
- ▶ The `value` attribute sets the initial value displayed in the form (Fig. 3.10).
- ▶ The spinner control includes only the valid numbers. (不是所有瀏覽器都有)
- ▶ If the user attempts to enter an invalid value by typing in the text field, a callout pointing to the number `input` element will instruct the user to enter a valid value.

3.1.8 `input` Type range

- ▶ The `range` input type appears as a *slider* control in Chrome, Safari and Opera (Fig. 3.12).
- ▶ You can set the minimum and maximum and specify a value.
- ▶ The range input type is *inherently self-validating* when it is rendered by the browser as a slider control, because *the user is unable to move the slider outside the bounds of the minimum or maximum value.*

3.1.9 `input` Type `search`

- ▶ The `search input type` provides a search field for entering a query.
- ▶ This `input` element is functionally equivalent to an `input` of type `text`.
- ▶ When the user begins to type in the search field, Chrome and Safari display **an X that can be clicked to clear the field** (Fig. 3.13).

3.1.10 `input` Type `tel`

- ▶ The `tel` input type enables the user to enter a telephone number—mobile browsers typically display a keypad specific to entering phone numbers for this input type.
- ▶ At the time of this writing, the `tel` input type is rendered as a text field in all of the browsers.
- ▶ HTML5 does *not* self validate the `tel` input type.
- ▶ To ensure that the user enters a phone number in a proper format, we've added a pattern attribute that uses a *regular expression* to determine whether the number is in the format:
 - (555) 555-5555
- ▶ When the user enters a phone number in the wrong format, a callout appears requesting the proper format, pointing to the `tel` input element (Fig. 3.14).

3.1.11 `input type="time"`

- ▶ The `time input type` enables the user to enter an hour, minute, seconds and fraction of second (Fig. 3.15).
- ▶ The HTML5 specification indicates that a time must have two digits representing the hour, followed by a colon (:) and two digits representing the minute.
- ▶ Optionally, you can also include a colon followed by two digits representing the seconds and a period followed by one or more digits representing a fraction of a second (shown as `ff` in our sample text to the right of the time input element in Fig. 3.15).

3.1.12 `input` Type `url`

- ▶ The `url` input type enables the user to enter a URL.
- ▶ The element is rendered as a text field, and the proper format is `http://www.deitel.com`.
- ▶ If the user enters an improperly formatted URL (e.g., `www.deitel.com` or `www.deitelcom`), the URL will *not* validate (Fig. 3.16).
- ▶ HTML5 does not check whether the URL entered is valid; rather it validates that the URL entered is in the proper format.

3.1.13 `input` Type `week`

- ▶ The `week input type` enables the user to select a year and week number in the format `yyyy-wnn`, where `nn` is `01–53`—for example, `2012-w01` represents the first week of 2012. Internet Explorer, Firefox and Safari render a text field.
- ▶ Chrome renders an up-down control.
- ▶ Opera renders *week control* with a down arrow that, when clicked, brings up a calendar for the current month with the corresponding week numbers listed down the left side.



3.2 input and dataList Elements and autoComplete Attribute

- ▶ Figure 3.17 shows how to use the new **autocomplete attribute** and **dataList element**.



```
1  <!DOCTYPE html>
2
3  <!-- Fig. 3.17: autocomplete.html -->
4  <!-- New HTML5 form autocomplete attribute and datalist element. -->
5  <html>
6      <head>
7          <meta charset="utf-8">
8          <title>New HTML5 autocomplete Attribute and datalist Element</title>
9      </head>
10
11     <body>
12         <h1>Autocomplete and Datalist Demo</h1>
13         <p>This form demonstrates the new HTML5 autocomplete attribute
14             and the datalist element.
15         </p>
16
17         <!-- turn autocomplete on -->
18         <form method = "post" autocomplete = "on">
19             <p><label>First Name:
20                 <input type = "text" id = "firstName"
21                     placeholder = "First name" /> (First name)
22             </label></p>
```

Fig. 3.17 | New HTML5 form autocomplete attribute and datalist element. (Part I of 6.)


```
23 <p><label>Last Name:
24   <input type = "text" id = "lastName"
25     placeholder = "Last name" /> (Last name)
26 </label></p>
27 <p><label>Email:
28   <input type = "email" id = "email"
29     placeholder = "name@domain.com" /> (name@domain.com)
30 </label></p>
31 <p><label for = "txtList">Birth Month:
32   <input type = "text" id = "txtList"
33     placeholder = "Select a month" list = "months" />
34   <datalist id = "months">
35     <option value = "January">
36     <option value = "February">
37     <option value = "March">
38     <option value = "April">
39     <option value = "May">
40     <option value = "June">
41     <option value = "July">
42     <option value = "August">
43     <option value = "September">
44     <option value = "October">
45     <option value = "November">
```

Fig. 3.17 | New HTML5 form autocomplete attribute and datalist element. (Part 2 of 6.)

```
46         <option value = "December">
47     </datalist>
48 </label></p>
49 <p><input type = "submit" value = "Submit" />
50     <input type = "reset" value = "Clear" /></p>
51 </form>
52 </body>
53 </html>
```

Fig. 3.17 | New HTML5 form autocomplete attribute and datalist element. (Part 3 of 6.)

Autocomplete and Datalist Demo

This form demonstrates the new HTML5 autocomplete attribute and the datalist element.

First Name: (First name)

Last Name: (Last name)

Email: (name@domain.com)

Birth Month:

Submit

Clear

January

June

July

3.2.1 `input` Element `autocomplete` Attribute

- ▶ The `autocomplete` attribute can be used on `input` types to automatically fill in the user's information based on previous input—such as name, address or e-mail.
- ▶ You can enable `autocomplete` for an entire form or just for specific elements.
- ▶ For example, an online order form might set `autocomplete = "on"` for the name and address inputs and set `autocomplete = "off"` for the credit card and password inputs for security purposes.



Error-Prevention Tip 3.1

The `autocomplete` attribute works only if you specify a `name` or `id` attribute for the `input` element.

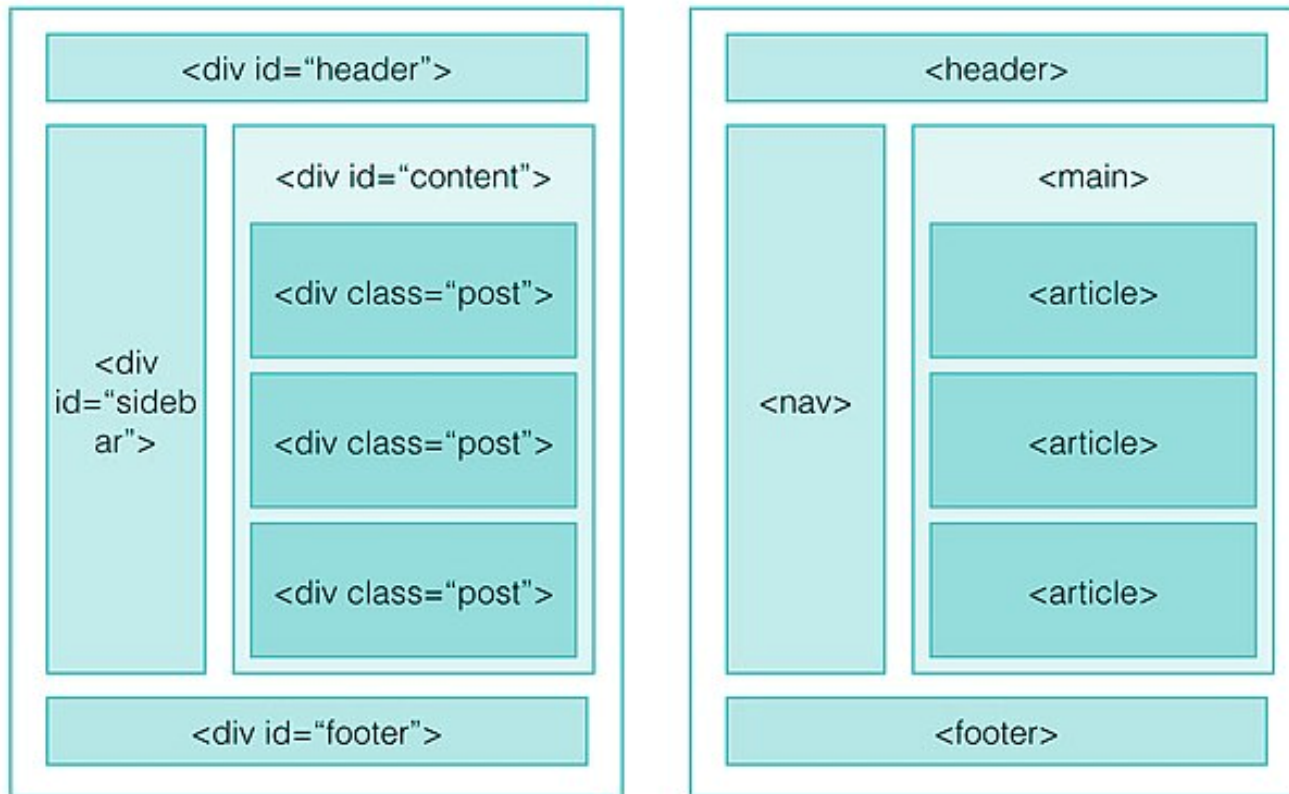
3.2.2 dataList Element

- ▶ The **datalist element** provides input options for a text input element.
- ▶ At the time of this writing, dataList support varies by browser.
- ▶ In this example, we use a dataList element to obtain the user's birth month.
- ▶ *Typical autocomplete example: Google search*

3.3 Page-Structure Elements

- ▶ HTML5 introduces several new page-structure elements (Fig. 3.18) that meaningfully identify areas of the page as headers, footers, articles, navigation areas, asides, figures and more.

Page-Structure Elements



https://www.w3schools.com/html/tryit.asp?filename=tryhtml_layout_float